THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS VI — MATHEMATICS · FULL-YEAR PRACTICE PAPER

Annual Examination Pattern · Session 2026-27 | CBSE / RBSE

Class: 6	Subject: Mathematics	Set 03	Time: 2 Hours 30 Minutes	Max. Marks: 60

General Instructions:

1. This question paper contains 16 questions. All questions are compulsory.
2. Question paper is divided into FIVE sections – Section 'A', 'B', 'C', 'D' and 'E'.
3. In Section 'A' question number 1 has 12 multiple choice sub-parts carrying 1 mark each.
4. In Section 'B' question numbers 2 to 7 are very short answer type questions carrying 2 marks each.
5. In Section 'C' question numbers 8 to 10 are short answer (SA) type questions carrying 3 marks each.
6. In Section 'D' question numbers 11 to 13 are long answer (LA) type questions carrying 5 marks each.
7. In Section 'E' question numbers 14 to 16 are case-based questions whose sub-parts are of 1, 1 and 2 marks respectively.
8. There is no overall choice. However, internal choice has been provided in 1 question of Section B, 1 question of Section C, 2 questions of Section D, and in the 2-mark sub-part of every case-based question in Section E. Attempt only one of the alternatives in all such questions.
9. Draw neat figures wherever required. Use of calculators is NOT allowed. Write the serial number of the question before attempting it.

SECTION A — Multiple Choice Questions (12 × 1 = 12) — 1 questions · 12 marks

1. Question 1 consists of multiple-choice questions (i–xii) of 1 mark each. Choose the correct option.

(i) In the sequence 1, 2, 4, 8, 16, ... (each term is double the previous one), the 9th term is: **1**

(a) 128 (b) 512 (c) 144 (d) 256

(ii) At 6 o'clock the hour hand and the minute hand of a clock form: **1**

(a) a right angle (b) a straight angle
(c) an obtuse angle (d) a reflex angle

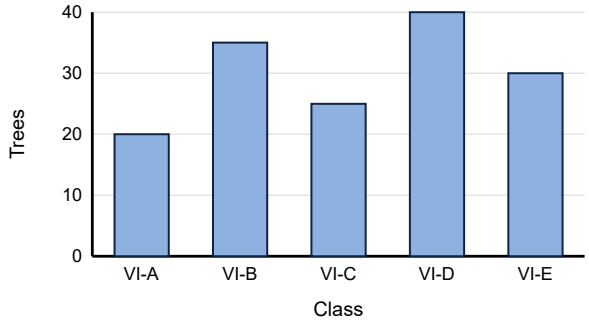
(iii) The number of terms (including 3 itself and 1) in the Collatz sequence that starts with 3 is: **1**

(a) 6 (b) 7 (c) 8 (d) 10

(iv) A tally chart shows 4 bundles of five and 3 single strokes. The number represented is: **1**

(a) 43 (b) 20 (c) 19 (d) 23

(v) The bar graph shows trees planted by five classes. The total number of trees planted is: **1**



(a) 120 (b) 150 (c) 160 (d) 130

(vi) The HCF of 24 and 36 is: **1**

(a) 6 (b) 12 (c) 24 (d) 72

(vii) Which of the following numbers is divisible by 12? **1**

(a) 124 (b) 138 (c) 150 (d) 132

(viii) Which fraction is equivalent to 18/24? **1**

(a) $\frac{3}{4}$ (b) $\frac{2}{3}$ (c) $\frac{5}{6}$ (d) $\frac{6}{7}$

(ix) The value of 3/4 − 2/5 is: **1**

(a) $\frac{7}{20}$ (b) $\frac{1}{10}$ (c) $\frac{1}{9}$ (d) $\frac{1}{4}$

(x) The perimeter of an equilateral triangle of side 9 cm is: **1**

(a) 18 cm (b) 81 cm (c) 27 cm (d) 36 cm

(xi) Which of the following integers is the greatest? **1**

(a) −3 (b) −15 (c) −9 (d) −21

(xii) The number of lines of symmetry of a square is: **1**

(a) 2 (b) 1 (c) 8 (d) 4

SECTION B — Very Short Answer (6 × 2 = 12) — 6 questions · 12 marks

Questions 2 to 7 carry 2 marks each.

2. Write tally marks for (i) 13 (ii) 22. (Use bundles of five.) **2**

3. Subtract: $4\frac{1}{3} - 1\frac{5}{6}$. **2**

4. Find: (i) (−12) + 7 (ii) 9 + (−15). **2**

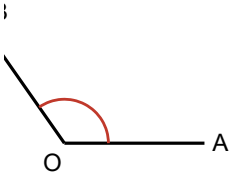
5. Write 84 as a product of prime factors. **2**

OR

Write 90 as a product of prime factors.

6. Write the next three terms of 2, 6, 18, 54, ... and state the rule. **2**

7. Measure the angle in the figure and write its type. **2**



SECTION C — Short Answer (3 × 3 = 9) — 3 questions · 9 marks

Questions 8 to 10 carry 3 marks each.

8. Meena asked 15 classmates their favourite colour: Red, Blue, Green, Blue, Red, Blue, Yellow, Green, Blue, Red, Blue, Green, Red, Blue, Yellow. Make a frequency table and tell which colour is the most popular. **3**

9. Ramu has $2\frac{1}{2}$ m of cloth. He uses 3/4 m for each of 3 cushion covers. How much cloth is left? **3**

OR

Maya reads 1/5 of a book on Monday and 3/10 on Tuesday. (i) What fraction has she read? (ii) How many pages remain if the book has 200 pages?

10. Take 57. Reverse its digits and add. Keep repeating with each new sum until you get a palindrome. Show every step. **3**

SECTION D — Long Answer (3 × 5 = 15) — 3 questions · 15 marks

Questions 11 to 13 carry 5 marks each.

11. (i) Write all the factors of 72. (ii) Write 72 and 108 as products of primes. (iii) Find the HCF and the LCM of 72 and 108. **5**

OR

(i) List all primes up to 50 using the Sieve of Eratosthenes (write the first four steps in words). (ii) Write all pairs of twin primes up to 50.

12. A teacher recorded the books read by 20 students in a month: 2, 3, 1, 2, 4, 2, 3, 1, 2, 5, 3, 2, 4, 1, 2, 3, 2, 5, 3, 2. (i) Make a frequency table. (ii) Draw a bar graph (1 cm = 1 student on the vertical axis). (iii) How many students read at least 4 books? **5**

OR

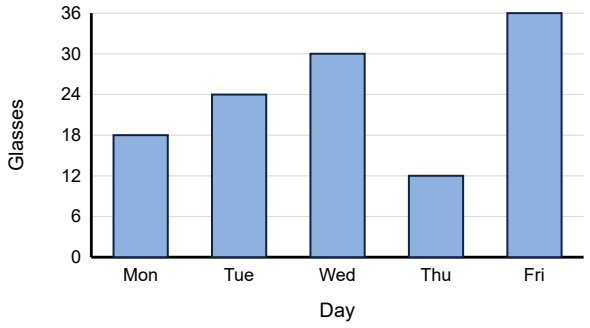
The number of cycles sold in a shop in six months is: Jan 25, Feb 40, Mar 30, Apr 15, May 35, Jun 20. (i) Draw a pictograph using 1 symbol = 5 cycles. (ii) In which month were the most cycles sold? (iii) Find the total cycles sold.

13. A farmer's rectangular field is 40 m long and 25 m wide. 2/5 of the field is used for wheat, 1/4 for mustard and the rest for vegetables. Find (i) the area of the field (ii) the area for wheat (iii) the area for mustard (iv) the area for vegetables (v) the fraction of the field used for vegetables. **5**

SECTION E — Case-Based Questions (3 × 4 = 12) — 3 questions · 12 marks

Questions 14 to 16 carry 4 marks each (1 + 1 + 2).

14. The school canteen records the number of glasses of lassi sold from Monday to Friday in a bar graph.



(i) On which day were the fewest glasses sold? 1

(ii) How many glasses were sold in the whole week? 1

(iii) How many more glasses were sold on Friday than on Monday? 2

OR

(iii) One glass costs ₹20. Find the money collected on Wednesday.

15. A submarine is at 35 m below sea level. We write this as −35 m. It rises 12 m to take photographs and then dives 20 m to study a coral reef.

(i) At what level is the submarine after rising 12 m? 1

(ii) At what level is it after diving 20 m? 1

(iii) How many metres is the final position below sea level? 2

OR

(iii) A bird flies at +18 m. Find the vertical distance between the bird and the submarine's final position.

16. Pooja designs a square floor tile for the school hall. She asks: which symmetries does a square tile have, and how does a plain rectangular (non-square) tile differ?

(i) Write the number of lines of symmetry of a square tile. 1

(ii) What is the order of rotational symmetry of a square tile? 1

(iii) Through how many degrees must a square tile be rotated (smallest turn) to look the same? 2

OR

(iii) Give the number of lines of symmetry and the order of rotational symmetry of a rectangular (non-square) tile.